OPEN SOURCE / LOCAL BIOINFORMATICS / AGENT ASSISTANCE

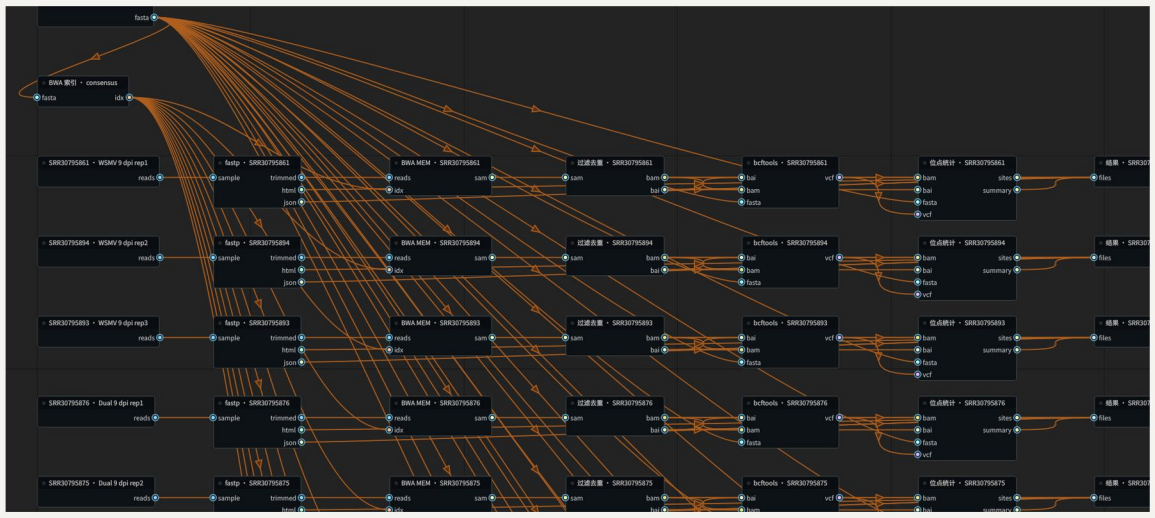
OpenBioWorkstation

本地生信工作流与 Agent 协作

Visual workflows and agent-assisted bioinformatics

NodeGraphQt / Snakemake / MCP

01 输入 → 工具 → 输出 / INPUT → TOOLS → OUTPUT



真实 WSMV 案例截图；包含案例专用节点。
Actual WSMV case screenshot; includes case-specific nodes.

02 过程检查与执行 / INSPECTION AND EXECUTION

端口格式检查 / Port checks

```
[界面] 已拒绝连线 fastp · SRR30795861.json → 位点统计 · SRR30795861.bam：输出为 JSON，输入需要 BAM
```

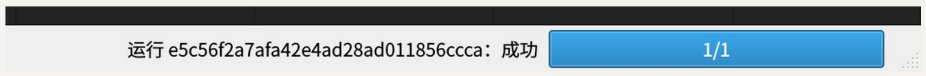
拒绝不兼容格式 / Reject incompatible formats

运行状态 / Execution state



显示节点状态 / Display node execution states

单步与缓存 / Reruns and cache



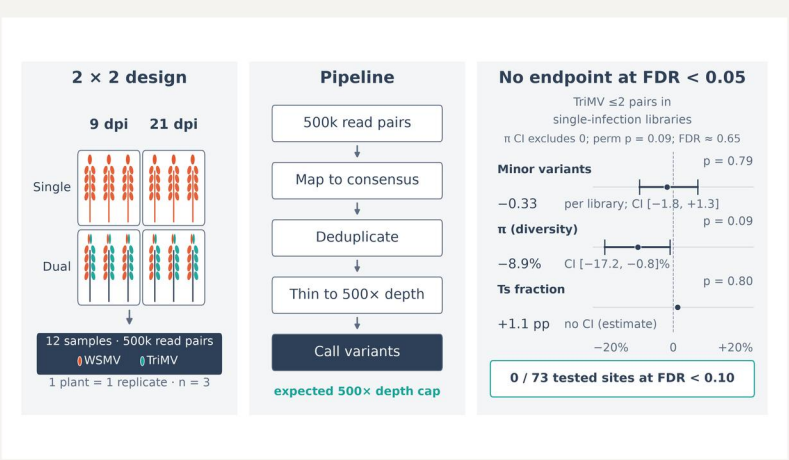
局部重跑 / Partial runs and cached skips

文件预览 / File previews



有限记录预览 / Bounded record inspection

03 研究案例 / RESEARCH CASE: WSMV × TriMV



12 个 RNA-seq 文库
12 RNA-seq libraries

0/73 位点达到 FDR < 0.10
0/73 sites at FDR < 0.10

未检出显著关联
No significant association detected

阴性结果不证明等效
A negative result is not equivalence

04 与 Agent 协作 / WORK WITH AN AGENT

研究者定义问题 → Agent 搭图 → dry-run → 执行与核对
Define the question → build → dry-run → execute → inspect

中英教程 / Tutorial: docs/agent-workflow.en.md

研究附件 / Research PDFs: promo/research/

Linux 单机；MCP stdio 已验证；具体 Agent 客户端需单独验证。
Single machine; stdio tested; validate each agent client separately.

github.com/ArchitektSpeer/openbioworkstation

GPL-3.0